

Ankur

Contingency-plan trigger engine for block-scale monsoon onset and break
SIH26086 · Hyperlocal Monsoon Onset & Break Prediction System (Block/Village Scale)
 Ministry of Earth Sciences · Miscellaneous · Software
 Vellore Institute of Technology, Vellore

1. Team details

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2. Problem

SIH26086 asks for a 7-to-30-day outlook of monsoon onset, break and revival at block / panchayat scale, plus crop advisories. That forecast already exists. In May 2026 IMD launched an AI block-level onset model (3,196 blocks, four weeks ahead, ~4-day error, five-day rain plus no prolonged dry spell over 30 days) and pushes it through AgriStack. BharatFS is operational at 6 km. Building another onset classifier is a worse copy of the organisation’s own system.

Two gaps remain. First, a probability is not a decision: “60% onset in week 3” does not tell a farmer with four acres and no borewell whether to sow on Tuesday. Re-sowing is the loss — in one Maharashtra soybean season about 20% of 43 lakh hectares needed re-sowing, including from poor moisture at sowing. Second, India already wrote the countermeasure: ~650 District Agriculture Contingency Plans (ICAR/CRIDA). The Sirsa (Haryana) plan, 2011, already maps “normal onset followed by 15–20 days dry spell after sowing” to named pearl-millet varieties, a re-sow rule, and an order that the block stage seed. Those plans sit in scanned PDFs, triggered in a pre-season meeting, used after drought is declared. CRIDA itself calls village-level implementation a real challenge. Nobody has made the plans machine-readable and wired them to the live block forecast.

A farmer sows after the first rain, the rain stops for twenty days, and he buys seed twice. The government already wrote what he should do instead. Ankur fires that rule before the seed goes in the ground, and tells the Block Agriculture Officer how many quintals of the named variety to stage this week.

3. Proposed solution

Ankur is a contingency-plan trigger engine. It does not declare onset (only IMD does). It classifies the live moisture state into the **exact condition vocabulary the DACP already uses**, then retrieves the pre-approved action with a page citation. No agronomy is generated.

ID	Objective	What it means
O1	Cite, never invent	Every action is a retrieved DACP row (district, soil, crop, stage, variety, actor) with PDF page. Silent if the plan is silent.
O2	Speak the plan’s labels	Outputs are plan inputs: delayed onset; normal onset + 15–20 day dry spell after sowing; mid-season break at flowering.
O3	Turn probability into action	Calibrated dry-spell probability → sow / wait N days / re-sow with variety X using the farmer’s cost–loss ratio (second seed bag vs waiting).

ID	Objective	What it means
O4	Pre-position seed	Hectares × trigger probability → quintals of the named short-duration variety the block should stage this week.
O5	Stay under IMD	Cite IMD / open ensembles. Audit trail of forecast snapshot, advice issued, rain that followed. Zero onset declarations.

3.1 What we will not claim

- We do not declare monsoon onset or break, and we do not compete with IMD's May 2026 model, BharatFS, or AgriStack as a weather product.
- We do not write new sowing advice. Stale 2011 varieties are flagged for KVK review, not rewritten.
- Hackathon data is openly licensed only (data.gov.in/CDSP gridded rainfall and normals, ECMWF open data, satellite precipitation). Production is described as an MoU / AgriStack API. Restricted IMD AWS feeds are not scraped.

4. Technical approach

Layer	What it does
A. Rule base	Parse DACPs (start: Haryana, flagship Sirsa) into (district, farming situation, soil, crop, aberration, stage, action, variety, seed rate, actor). Layout-aware extraction + schema-constrained structured output, page citation, confidence flag. Retrieval only.
B. GIS / names	Join 2011 district plans to current block boundaries; soil-based sub-district split; crop and variety aliases (HHB-67 Improved vs HHB 67 imp). Reorganisation test: plans still filed under old state names.
C. Trigger	Daily root-zone water balance per block: observed rain + calibrated ensemble + available water capacity → moisture state → DACP condition. 1:1 map onto a pre-approved action.
D. Two users	Farmer: one Hindi sentence (sow / wait N days / re-sow with X) + reason + citation. Block officer: seed-demand nowcast in quintals. Query path is SELECT on precomputed triggers.

4.1 Machine learning (on top of IMD, not instead of it)

Component	Method	Reported metric
Ensemble calibration	Quantile mapping / gradient-boosted post-process of free ensembles to block dry-spell (≥ 5 day) probability.	Brier Skill Score vs climatology; reliability diagram
Cost-loss decision	Same p(dry spell), different sow/wait thresholds from the farmer's own re-sow cost vs waiting loss.	Economic value across cost-loss ratios
Extraction QA	Schema validation; F1 on a hand-annotated sample of plan rows. Low-confidence rows never SMS.	Field-level P/R/F1

Preprocessing (only this from our load-forecasting research): chronological sort and dedup, regular daily frequency with explicit gaps, short-gap forward-fill (≤ 2 days), 4 σ winsorization, linear interpolate, causal shifted lags and rolling windows, train-only scaler, last monsoon season held out. Retargeted from 15-minute load to daily rainfall and soil-moisture features. ENSO / IOD / MJO enter as lagged exogenous columns because the PS requires them. We do not import that paper's model zoo or MAPE ranking.

4.2 Stack

Plane	Job	Stack
Ingest	Open rainfall, ensembles, DACP PDFs. Licence tag on every source row.	Python; object store; Postgres + PostGIS
Decide	Nightly / replay water balance → condition → matched rows → seed demand. Audit log.	Python jobs; durable workflow if SMS must not double-fire
Serve	Farmer card + BAO map. Hindi template. Fallback JSON if DB dies mid-demo.	FastAPI, React, SMS gateway stub

5. Expected outcomes and demo

Finale demo: replay monsoon 2025 on ~10 blocks of Sirsa, Haryana (early rain, then a long dry spell). Naive rule “it rained → sow” says GO. Ankur says WAIT once calibrated dry-spell probability crosses the cost-loss threshold. The spell arrives. The matching DACP row lights up (variety + re-sow rule + PDF page). The officer panel shows quintals to have staged. Database kill still serves last triggers from fallback JSON. The product never prints an onset date.

Metric	Target
Brier Skill Score on block dry-spell vs climatology	Positive BSS on 2025 hold-out; reliability diagram shown
Lead time on triggers that match a DACP condition	Median ≥ 7 days before the spell is observed in replay

Metric	Target
Extraction F1 (condition + variety + action) on annotated Sirsa rows	≥ 0.85 ; low-confidence rows never auto-send
Decision value	Avoided re-sow hectares $\times$ seed cost/ha, reported as a range over cost–loss ratios
Onset dates declared by Ankur	Zero

5.1 Pilot (if adopted)

One district, MoU-shaped IMD/AgriStack feed, one KVK agronomist on the review queue, Hindi SMS to a consented farmer panel, BAO seed sheet used in the pre-season contingency meeting instead of a printed PDF. Pattern is copyable to N districts; this is not a 28-state weather app.

6. Budget

Software edition. No GPUs. Hackathon: existing laptops, ~₹10–15,000. Figures below are a **6-month single-district pilot** after SIH, not a national rollout.

Head	Use	INR
Cloud + storage	2 VMs, 300 GB PDFs/forecast snapshots, backups	1,00,000
Rule-base work	Gold annotation of Haryana DACPs; agronomist 8 half-days	1,40,000
GIS	Block shapefiles, 2011 overlay, soil join for Sirsa	70,000
SMS panel	Hindi transactional SMS, ~500 consented numbers	35,000
Field / KVK	Two trips: DAO + KVK, one pre-season meeting	50,000
Compliance	Licence memo, DPDP note, audit log, VAPT lite	50,000
Contingency 15%	Extra annotation / VM / later paid IMD extract if approved	67,000
Total	Single-district software pilot	5,12,000

Not budgeted: radars, drones, satellite tasking, a 28-language consumer app, or a foundation monsoon model.

7. Risks and compliance

Risk	Mitigation
Heard as an IMD clone	First claim on every slide: we do not forecast onset. Demo never prints an onset date.
IMD data-supply terms / restricted AWS	Named open licences on the demo. Production = MoU. No station files on GitHub.
Extractor invents a variety or seed rate	Page citation required to emit. F1 gate. Review queue. Agronomist on the gold set.
2011 varieties withdrawn	Show original text, flag stale, officer sees “review with KVK”.
Farmer loss after WAIT/GO	Decision support, IMD cited, probability shown, audit stored, DPDP consent. Not a substitute for the block officer.

8. References (selected)

- SIH26086, sih.gov.in — MoES; onset/break/revival; ENSO–IOD–MJO; maps; advisories; SMS.
- IMD AI monsoon advance model, May 2026; AgriStack / MoA dissemination. Block rainfall monitoring ~7,200 blocks (Aug 2025). BharatFS 6 km (May 2025, MoES).
- ICAR–CRIDA District Agriculture Contingency Plans; Sirsa, Haryana (CCS HAU / CRIDA, 30-06-2011).
- Maharashtra soybean re-sowing estimates (SOPA / state, cited season). IMD data-supply undertaking. Marathwada sowing-loss complaints against forecast use (public reporting, 2020s).